

Tutorial for week 13

Tan Junjie 12332175@mail.sustech.edu.cn

AM modulation $\rightarrow y(t) = x(t) \cos(\omega_c t)$

AM demodulation $\rightarrow w(t) = y(t) \cos(\omega_c t) = x(t) \cos^2(\omega_c t)$

$$= x(t) \left[\frac{1}{2} + \frac{1}{2} \cos(2\omega_c t) \right]$$

Convolution

Multiplication

$$x(t) * y(t)$$

$$x(t)y(t)$$

$\mathcal{A}$

$$X(j\omega)Y(j\omega)$$

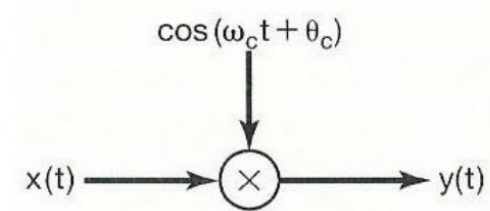
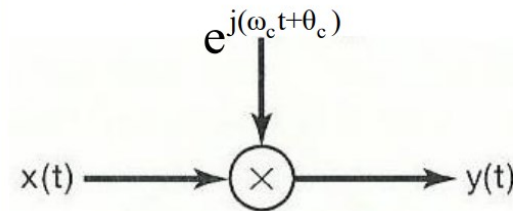
$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$$

$$= \frac{1}{2} x(t) + \frac{1}{2} x(t) \cos(2\omega_c t)$$

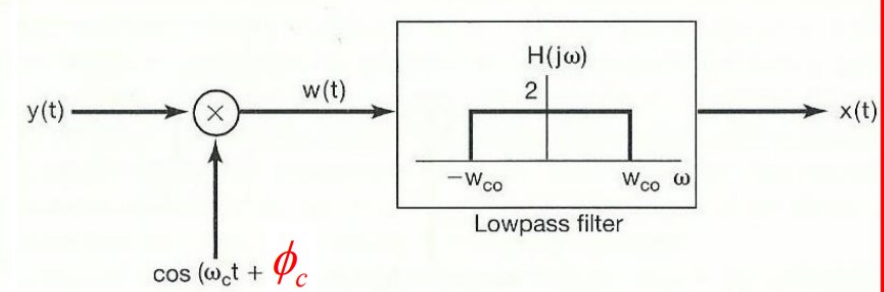
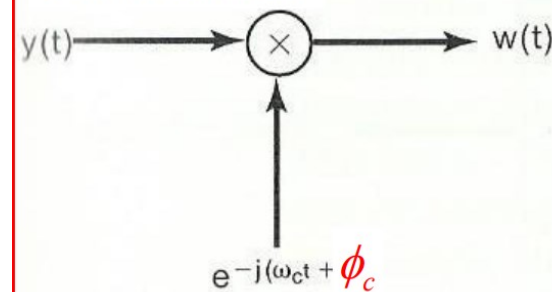
Complex exponential carrier:

Sinusoidal carrier:

AM modulation



De-modulation



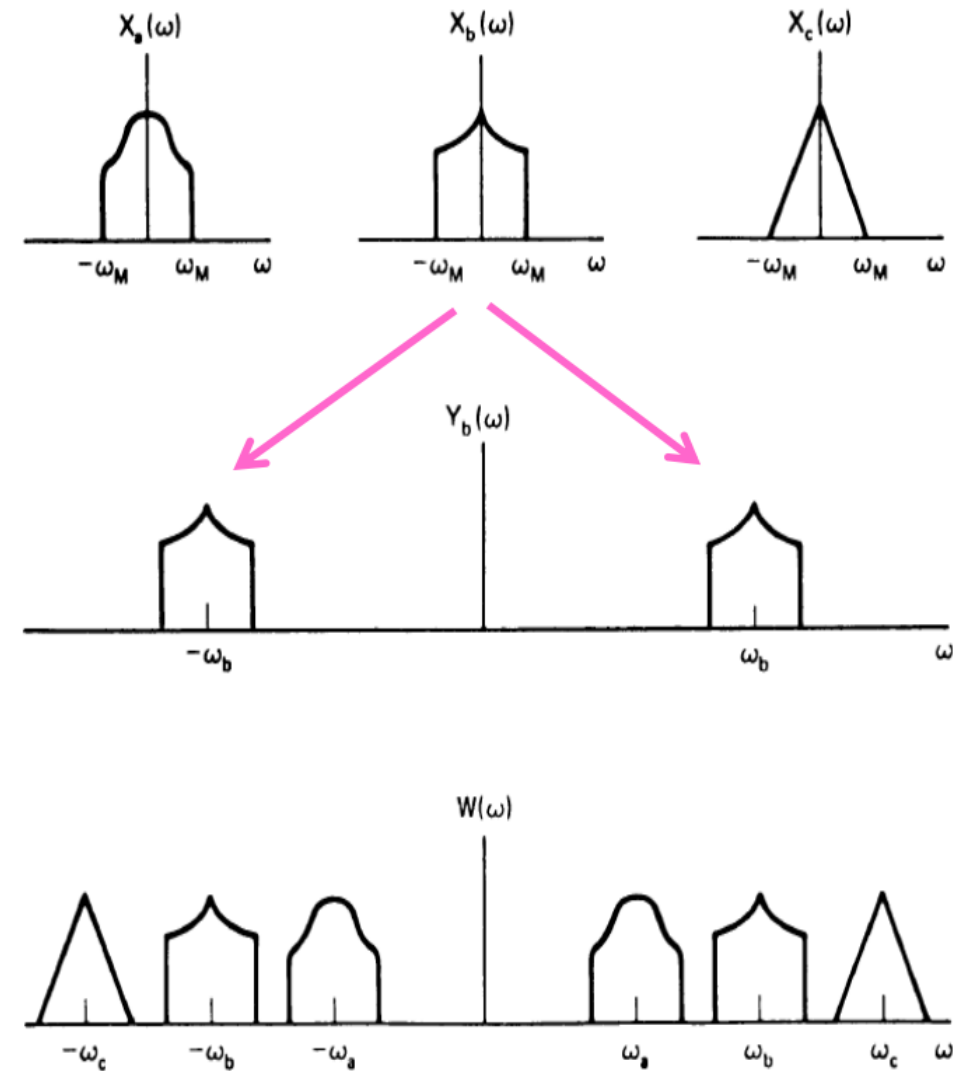
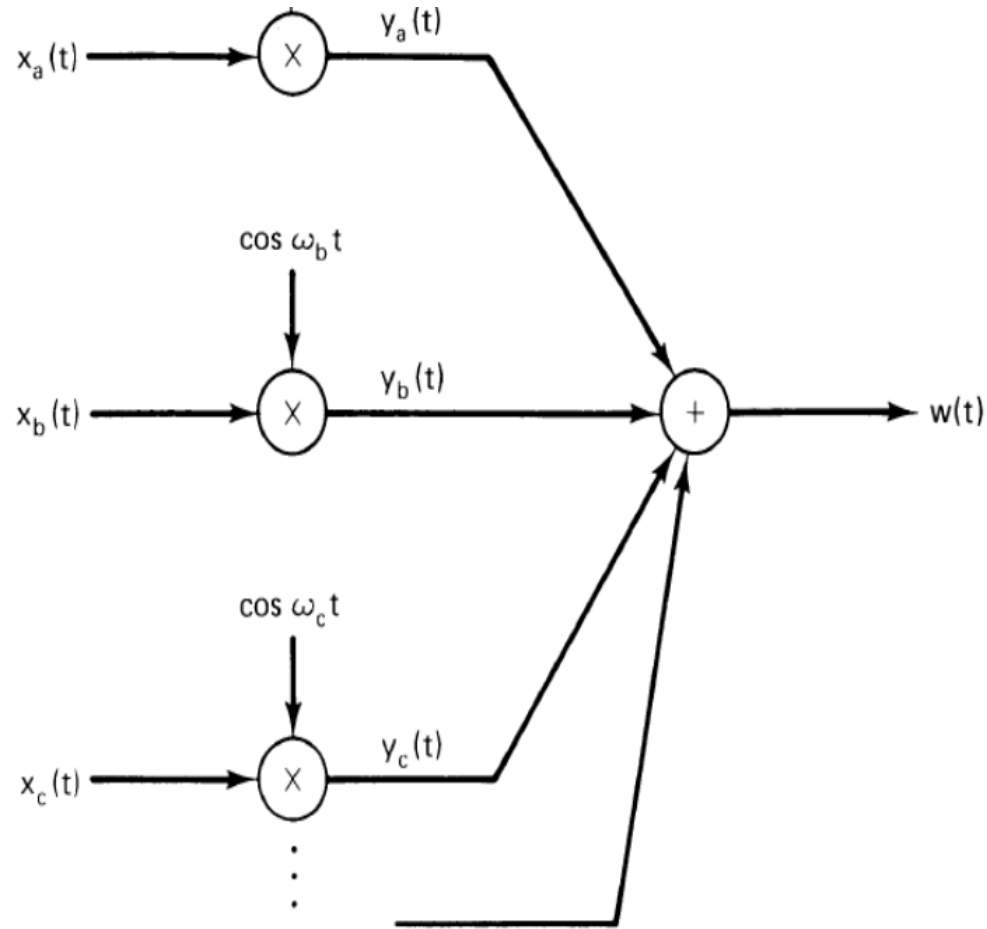
$$y(t) = e^{j(\omega_c t + \theta_c)} x(t)$$

$$w(t) = e^{-j(\omega_c t + \phi_c)} y(t) \\ = e^{j(\theta_c - \phi_c)} x(t)$$

$$w(t) = x(t) \cos(\omega_c t + \theta_c) \cos(\omega_c t + \phi_c)$$

$$= \frac{1}{2} \cos(\theta_c - \phi_c) x(t) + \frac{1}{2} x(t) \cos(2\omega_c t + \theta_c + \phi_c)$$

- Systems for transmitting signals provide more bandwidth than is required for one signal.
 - E.g., speech signal $\rightarrow 20 \sim 20 \text{ kHz}$
 - microwave channel $\rightarrow 300 \text{ MHz} \sim 300 \text{ GHz}$
 - satellite link $\rightarrow \text{a few hundred MHz} \sim 40 \text{ GHz}$
 (more in Fig. 8.18)



8.34. In discussing amplitude modulation systems, modulation and demodulation were carried out through the use of a multiplier. Since multipliers are often difficult to implement, many practical systems use a nonlinear element. In this problem, we illustrate the basic concept.

In Figure P8.34, we show one such nonlinear system for amplitude modulation. The system consists of squaring the *sum* of the modulating signal and the carrier and then bandpass filtering to obtain the amplitude-modulated signal.

Assume that $x(t)$ is band limited, so that $X(j\omega) = 0$, $|\omega| > \omega_M$. Determine the bandpass filter parameters A , ω_l , and ω_h such that $y(t)$ is an amplitude-modulated version of $x(t)$ [i.e., such that $y(t) = x(t) \cos \omega_c t$]. Specify the necessary constraints, if any, on ω_c and ω_M .

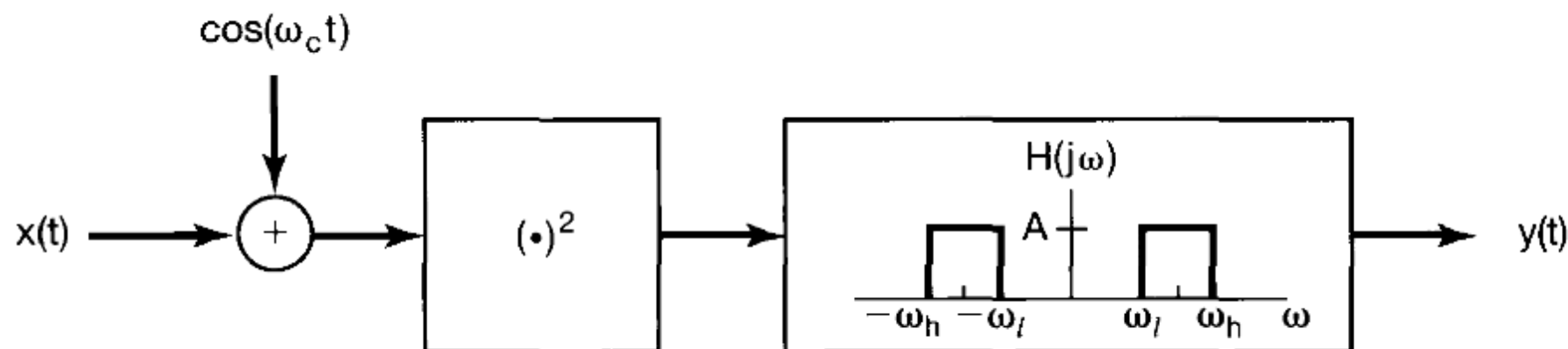


Figure P8.34

The output of the squarer is:

$$r(t) = [x(t) + \cos(\omega_c t)]^2 = x^2(t) + \cos^2(\omega_c t) + 2x(t) \cos(\omega_c t)$$

The bandpass filter should reject $x^2(t) + \cos^2(\omega_c t)$ and multiply the remainder by $1/2$. Therefore, $A = 1/2$. Since the spectral contribution of $2x(t) \cos \omega_c t$ is in the range $\omega_c - \omega_M \leq |\omega| \leq \omega_c + \omega_M$, we require $\omega_l = \omega_c - \omega_M$ and $\omega_h = \omega_c + \omega_M$.

Note that (i) the spectral contribution of $x^2(t)$ is in the range $|\omega| \leq 2\omega_M$ and (ii) the spectral contribution of $\cos^2(\omega_c t)$ is at $\omega = 0$ and $\omega = \pm 2\omega_c$. Therefore, we need to ensure that

$$\omega_l > 2\omega_M \implies \omega_M < \omega_c/3$$

We also need to ensure that

$$\omega_h < 2\omega_c \implies \omega_M < \omega_c$$

8.35. The modulation-demodulation scheme proposed in this problem is similar to sinusoidal amplitude modulation, except that the demodulation is done with a square wave with the same zero-crossings as $\cos \omega_c t$. The system is shown in Figure P8.35(a); the relation between $\cos \omega_c t$ and $p(t)$ is shown in Figure P8.35(b). Let the input signal $x(t)$ be a band-limited signal with maximum frequency $\omega_M < \omega_c$, as shown in Figure P8.35(c).

- (a) Sketch and dimension the real and imaginary parts of $Z(j\omega)$, $P(j\omega)$, and $Y(j\omega)$, the Fourier transforms of $z(t)$, $p(t)$, and $y(t)$, respectively.
- (b) Sketch and dimension a filter $H(j\omega)$ so that $v(t) = x(t)$.

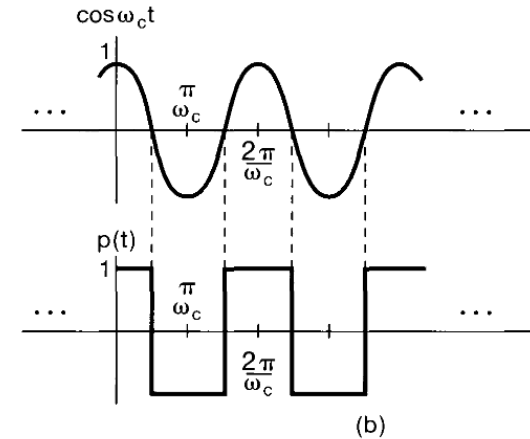
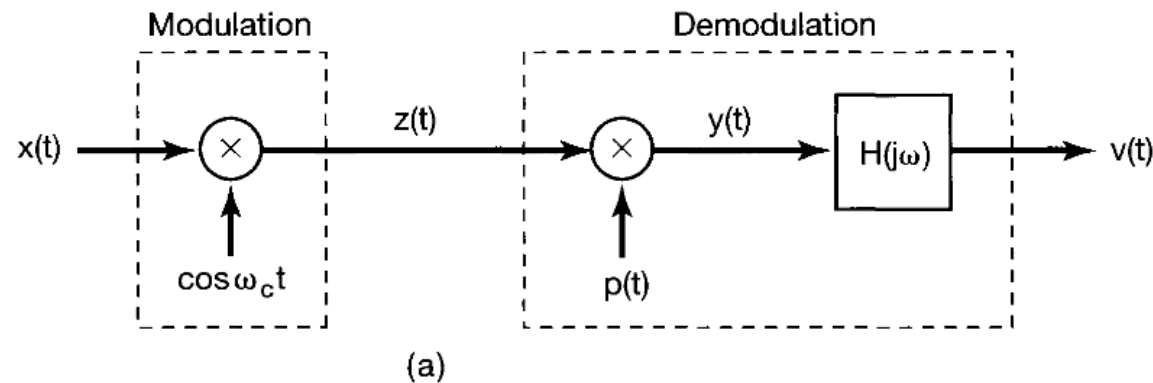
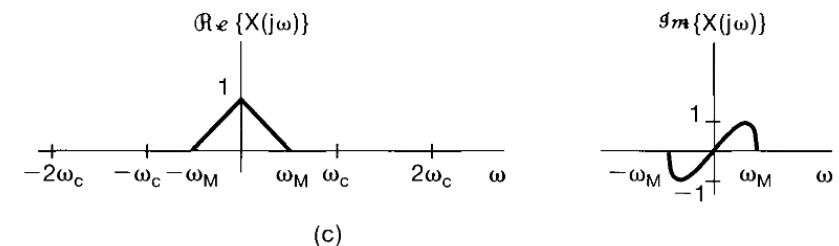


Figure P8.35



(a)

$$z(t) = x(t) \cos(\omega_c t) \xrightarrow{FT} Z(j\omega) = \frac{1}{2}X(j(\omega - \omega_c)) + \frac{1}{2}X(j(\omega + \omega_c))$$

The Fourier series coefficients

$$a_k = \frac{1}{T} \int_T p(t) e^{-jk \frac{2\pi}{T} t} dt \quad (1)$$

$$= \frac{\omega_c}{2\pi} \int_0^{2\pi/\omega_c} p(t) e^{-jk\omega_c t} dt \quad (2)$$

$$= \frac{\omega_c}{2\pi} \left(\int_0^{\pi/2\omega_c} e^{-jk\omega_c t} dt - \int_{\pi/2\omega_c}^{3\pi/2\omega_c} e^{-jk\omega_c t} dt + \int_{3\pi/2\omega_c}^{2\pi/\omega_c} e^{-jk\omega_c t} dt \right) \quad (3)$$

$$a_k = \frac{4 \sin(k\pi/2)}{k} (k \neq 0)$$

$$a_0 = 0$$

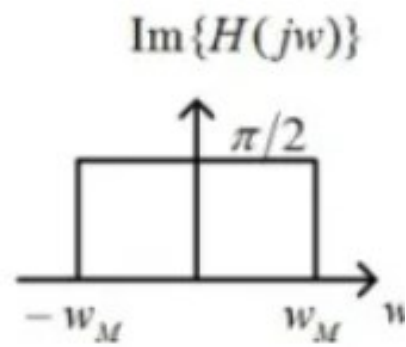
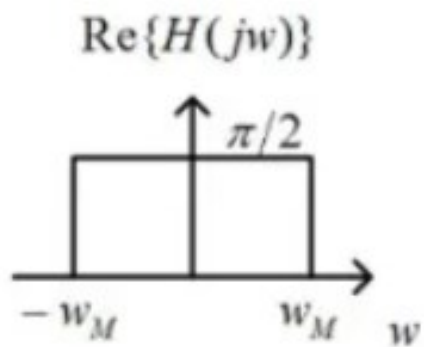
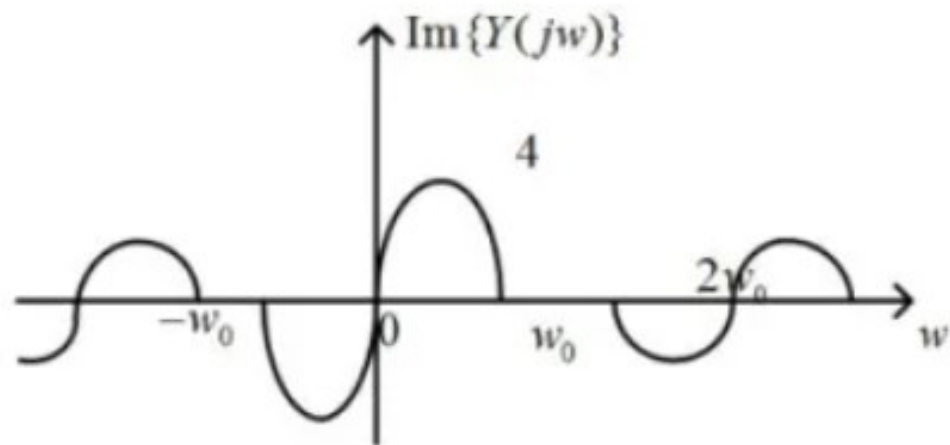
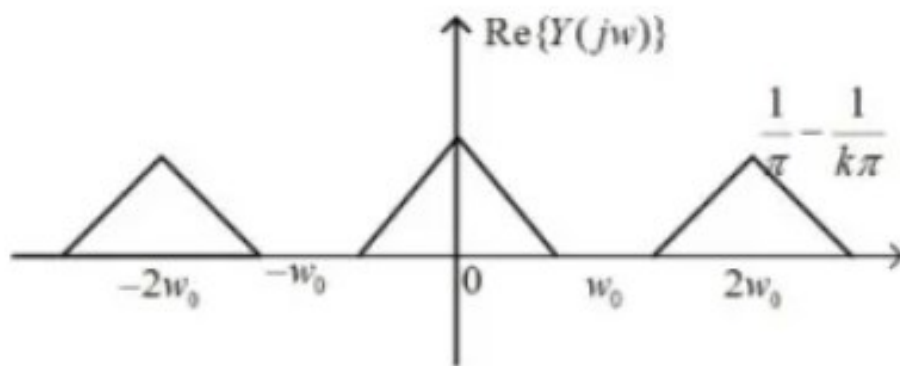
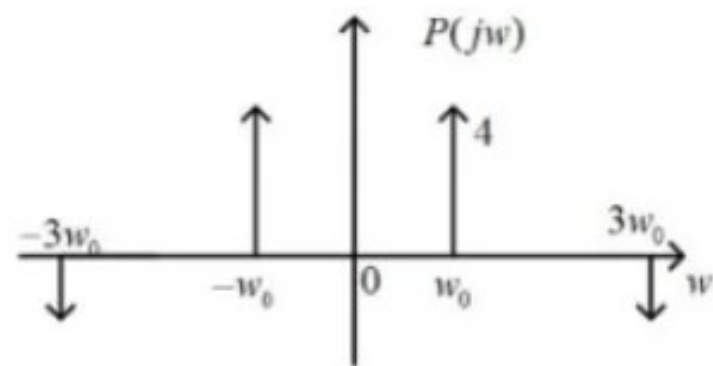
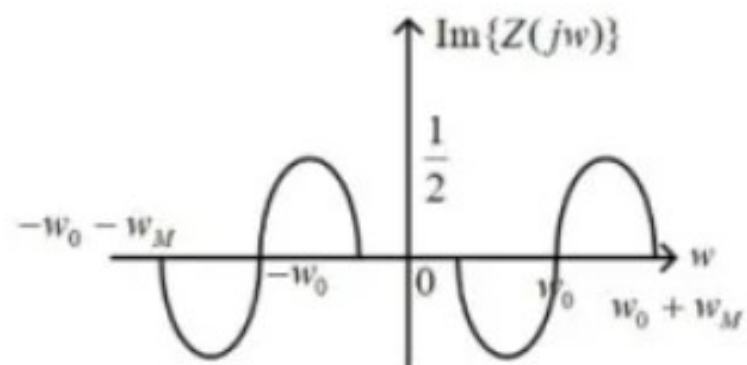
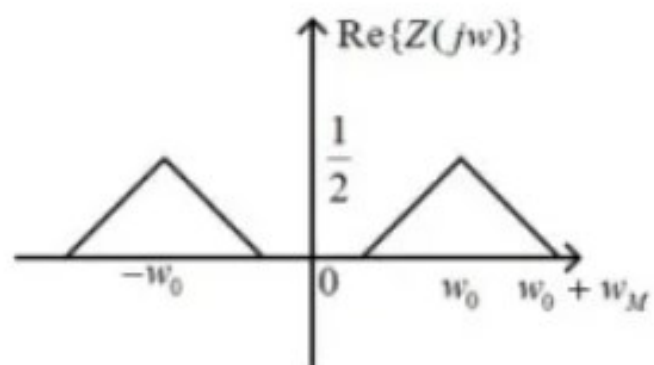
Also, $p(t)$ can be viewed as twice the standard periodic square wave and then minus one. Then use the table.

$$p(t) = 2x(t) - 1$$

$$P(j\omega) = \sum_{k=-\infty, k \neq 0}^{+\infty} \frac{4 \sin(k\pi/2)}{k} \delta(\omega - k\omega_0)$$

Since $y(t) = z(t)y(t)$

$$Y(j\omega) = \frac{1}{2\pi} [Z(j\omega) * P(j\omega)]$$



8.38. In Figure P8.38(a), a communication system is shown that transmits a band-limited signal $x(t)$ as periodic bursts of high-frequency energy. Assume that $X(j\omega) = 0$ for $|\omega| > \omega_M$. Two possible choices, $m_1(t)$ and $m_2(t)$, are considered for the modulating signal $m(t)$. $m_1(t)$ is a periodic train of sinusoidal pulses, each of duration D , as shown in Figure P8.38(b). That is,

$$m_1(t) = \sum_{k=-\infty}^{\infty} p(t - kT),$$

where

$$p(t) = \begin{cases} \cos \omega_c t, & |t| < (D/2) \\ 0, & |t| > (D/2) \end{cases}$$

$m_2(t)$ is $\cos \omega_c t$ periodically blanked or gated; that is, $m_2(t) = g(t) \cos \omega_c t$, where $g(t)$ is as shown in Figure P8.38(b).

The following relationships between the parameters T , D , ω_c , and ω_M are assumed:

$$D < T,$$

$$\omega_c \gg \frac{2\pi}{D},$$

$$\frac{2\pi}{T} > 2\omega_M.$$

Also, assume that $[\sin(x)]/x$ is negligible for $x \gg 1$.

Determine whether, for some choice of ω_{lp} , either $m_1(t)$ or $m_2(t)$ will result in a demodulated signal $x(t)$. For each case in which your answer is yes, determine an acceptable range for ω_{lp} .

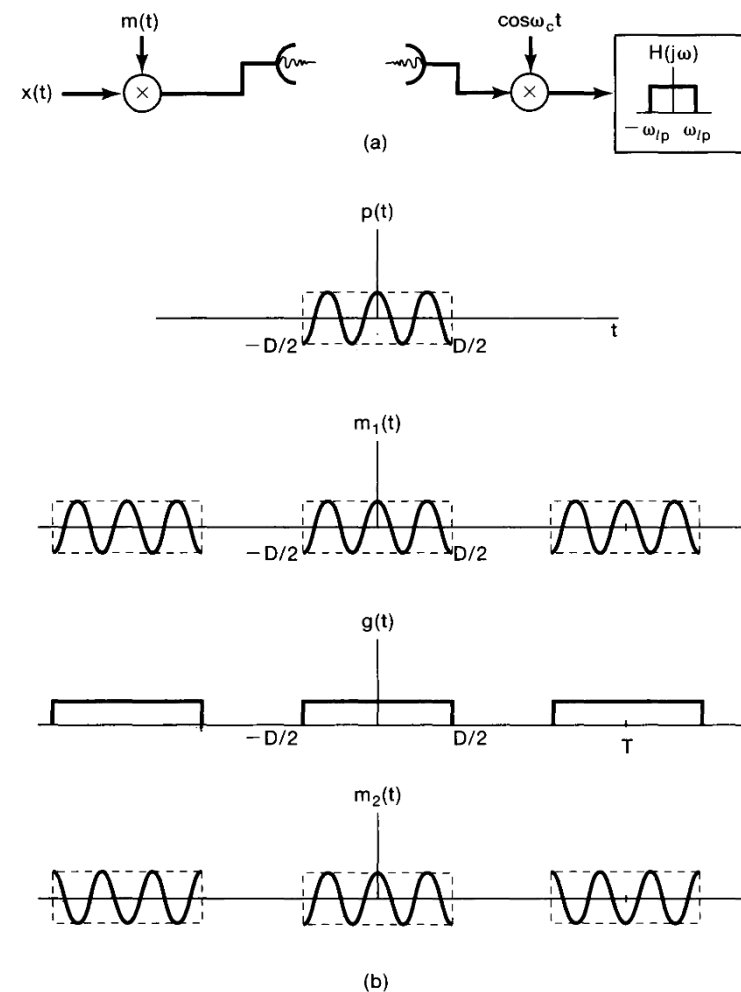


Figure P8.38

One of the key issues to note in this problem is that the structure of the demodulator is that of a synchronous demodulator. Therefore, the input signal to the demodulator has to have a replica of $X(j\omega)$ centered around ω_c . Only then will the demodulator be successful in recovering $x(t)$.

Case 1:

$$P(j\omega) = \frac{\sin [(\omega - \omega_c)D/2]}{\omega - \omega_c} + \frac{\sin [(\omega + \omega_c)D/2]}{\omega + \omega_c}$$

$M_1(j\omega)$ will consist of impulses which occur at intervals of $2\pi/T$ weighted replicas of $P(j\omega)$.

$$M_1(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} P(j(\omega - \frac{2\pi n}{T}))\delta(\omega - \frac{2\pi n}{T})$$

Furthermore, note that if $y_1(t) = x(t)m_1(t)$, then we have

$$Y_1(j\omega) = \frac{1}{2\pi}[X(j\omega) * M_1(j\omega)]$$

Therefore, $Y_1(j\omega)$ will consist of weighted replicas of $X(j\omega)$ which occur every $2\pi/T$. Note that unless ω_c is a multiple of $2\pi/T$, $M_1(j\omega) = 0$ for $\omega = \pm\omega_c$. If $2\pi/T$ is arbitrary, (i.e., it is not specified to be a multiple of ω_c) $Y_1(j\omega)$ has no replicas of $X(j\omega)$ centered around ω_c . Since $y_1(t)$ constitutes the input to the demodulator, the signal $r(t)$ at the output of the demodulator will not be proportional to $x(t)$.

Case 2:

$$M_2(j\omega) = \frac{1}{2}G(j(\omega - \omega_c)) + \frac{1}{2}G(j(\omega + \omega_c))$$

where

$$G(j\omega) = \sum_{k=-\infty}^{+\infty} \frac{2 \sin(k\pi D/T)}{k} \delta(\omega - 2\pi k/T)$$

Then

$$M_2(j\omega) = \sum_{k=-\infty}^{+\infty} \left[\frac{\sin(k\pi D/T)}{k} \delta(\omega - \omega_c - 2\pi k/T) + \sum_{k=-\infty}^{+\infty} \frac{\sin(k\pi D/T)}{k} \delta(\omega + \omega_c - 2\pi k/T) \right]$$

Clearly, $M_2(j\omega)$ has equal-valued impulses at $\pm\omega_c$. Therefore, the Fourier transform $Y_2(j\omega)$ of the signal $y_2(t) = x(t)m_2(t)$ has replicas of $X(j\omega)$ at $\pm\omega_c$. These replicas do not alias with other replicas of $X(j\omega)$ in $Y_2(j\omega)$ because $2\pi/T > 2\omega_M$. Thus, when demodulation is performed on $y_2(t)$, then $r(t)$ can be made proportional to $x(t)$ provided $\omega_M < \omega_{lp}$.

$$M_1(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} P(j(\omega - \frac{2\pi k}{T})) \delta(\omega - \frac{2\pi k}{T})$$

$$M_2(j\omega) = \sum_{k=-\infty}^{+\infty} \left[\frac{\sin(k\pi D/T)}{k} \delta(\omega - \omega_c - 2\pi k/T) + \sum_{k=-\infty}^{+\infty} \frac{\sin(k\pi D/T)}{k} \delta(\omega + \omega_c - 2\pi k/T) \right]$$